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a) Criação da topologia:

```
sudo python3 codigo.py
```

b) Visualizar os nós, conexões e interfaces

```
mininet> nodes
```

available nodes are:

h1 h2 h3 h4 h5 s1 s2 s3

```
mininet> net
```

h1 h1-eth0:s1-eth1

h2 h2-eth0:s1-eth2

h3 h3-eth0:s2-eth3

h4 h4-eth0:s3-eth2

h5 h5-eth0:s3-eth3

s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:s2-eth1

s2 lo: s2-eth1:s1-eth3 s2-eth2:s3-eth1 s2-eth3:h3-eth0

s3 lo: s3-eth1:s2-eth2 s3-eth2:h4-eth0 s3-eth3:h5-eth0

```
mininet> dump
```

<Host h1: h1-eth0:10.0.0.1 pid=109655>

<Host h2: h2-eth0:10.0.0.2 pid=109657>

<Host h3: h3-eth0:10.0.0.3 pid=109659>

<Host h4: h4-eth0:10.0.0.4 pid=109661>

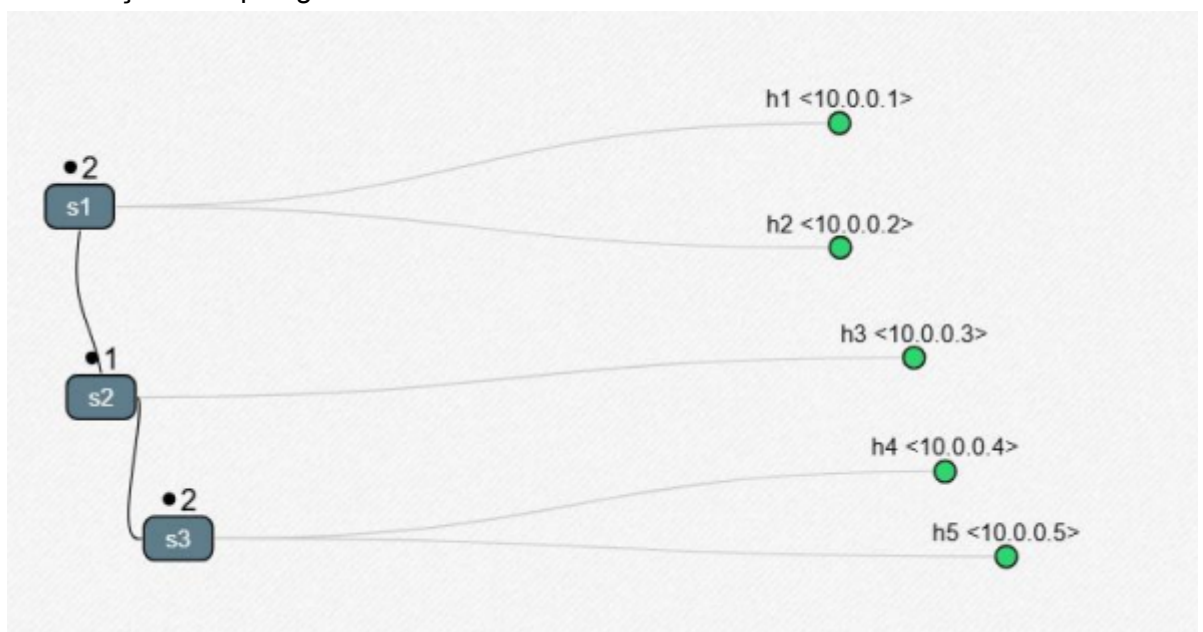
<Host h5: h5-eth0:10.0.0.5 pid=109663>

<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None pid=109668>

<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=109671>

<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=109674>

c) Visualização da topologia:



d) Teste de ping:

```
mininet> pingall
```

```
*** Ping: testing ping reachability
```

```
h1 -> h2 h3 h4 h5
```

```
h2 -> h1 h3 h4 h5
```

```
h3 -> h1 h2 h4 h5
```

```
h4 -> h1 h2 h3 h5
```

```
h5 -> h1 h2 h3 h4
```

```
*** Results: 0% dropped (20/20 received)
```

e) Regras baseadas em endereços MAC para alguns nós:

```
mininet> h1 ifconfig
```

```
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
```

```
inet 10.0.0.1 netmask 255.255.255.0 broadcast 10.0.0.255
```

```
inet6 fe80::70ed:c7ff:fe47:7416 prefixlen 64 scopeid 0x20<link>
```

```
ether 72:ed:c7:47:74:16 txqueuelen 1000 (Ethernet)
```

```
RX packets 31 bytes 4642 (4.5 KiB)
```

```
RX errors 0 dropped 0 overruns 0 frame 0
```

```
TX packets 25 bytes 1510 (1.4 KiB)
```

```
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
```

```
inet 127.0.0.1 netmask 255.0.0.0
```

```
inet6 ::1 prefixlen 128 scopeid 0x10<host>
```

```
loop txqueuelen 1000 (Loopback Local)
```

```
RX packets 4 bytes 448 (448.0 B)
```

```
RX errors 0 dropped 0 overruns 0 frame 0
```

```
TX packets 4 bytes 448 (448.0 B)
```

```
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
mininet> h5 ifconfig
```

```
h5-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
```

```
inet 10.0.0.5 netmask 255.255.255.0 broadcast 10.0.0.255
```

```
inet6 fe80::38b9:2fff:fe43:1b7 prefixlen 64 scopeid 0x20<link>
```

```
ether 3a:b9:2f:43:01:b7 txqueuelen 1000 (Ethernet)
```

```
RX packets 31 bytes 4642 (4.5 KiB)
```

```
RX errors 0 dropped 0 overruns 0 frame 0
```

```
TX packets 25 bytes 1510 (1.4 KiB)
```

```
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
```

```
inet 127.0.0.1 netmask 255.0.0.0
```

```
inet6 ::1 prefixlen 128 scopeid 0x10<host>
```

```
loop txqueuelen 1000 (Loopback Local)
```

```
RX packets 4 bytes 448 (448.0 B)
```

```
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 4 bytes 448 (448.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
mininet> sh ovs-ofctl add-flow s1 "dl_src=72:ed:c7:47:74:16,actions=normal"
mininet> sh ovs-ofctl add-flow s3 "dl_dst=3a:b9:2f:43:01:b7,actions=normal"
```

- f) Testes de ping para demonstrar que as regras foram bem implementadas:

```
mininet> h1 ping -c 3 h5
PING 10.0.0.5 (10.0.0.5) 56(84) bytes of data.
64 bytes from 10.0.0.5: icmp_seq=1 ttl=64 time=1.23 ms
64 bytes from 10.0.0.5: icmp_seq=2 ttl=64 time=1.10 ms
64 bytes from 10.0.0.5: icmp_seq=3 ttl=64 time=1.09 ms

--- 10.0.0.5 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 1.09/1.14/1.23/0.06 ms
```